UNIVERSITÄT BAYREUTH

Economics VII: Chair of
Quantitative Economic History

Summer Term 2025 **Bayreuth, 28.04.2025**

Data Science in Economic History: Spatial Analysis and Mapping

Timur Öztürk

Tuesdays, 14:15-15:45, S56 (RW I / PC-Pool)

Course Summary

This Master's course provides an overview of spatial analysis and mapping in R with applications in economic history. The first part of the course introduces students to the basic toolkit in geospatial research (as used in economics). We will discuss the handling of vector and raster data, the processing of spatial data, map making, and a selection of more advanced topics. The second part of the course then applies the tools by replicating spatial analyses of research papers in economic history and long-run development. By the end, students will complete a project demonstrating their ability to apply spatial analysis methods to a historical economic research question.

General Information

The course will be taught in **English.**

Target Groups:

- Master's students in History & Economics, Philosophy & Economics, Economics, or related programmes.
- Advanced undergraduate students with a profound interest in econometrics and R.

Prerequisite: Students should be familiar with basic econometric methods. Prior exposure to R is recommended (although not strictly necessary) as we will not learn R from scratch.

Enrolment is restricted to 15 students. Please sign up for the course on CMLife (first come, first serve). Registration opens on 16.03.2025 and closes on 28.04.2025.

I will make slides, papers, and other resources available via e-learning.

Grading

- Two take-home assignments: 25% (12.5% each), due 27.05. and 24.06.2025.
- Final Project: 25% for the Presentation, 50% for the final submission due 30.09.2025.

Prof. Dr. Sebastian T. Braun I Economics VII: Chair of Quantitative Economic History, Universitätsstr. 30, 95447 Bayreuth
Tel: 0921 55-6256, Fax: 0921 55-84 6241, sebastian.braun@uni-bayreuth.de
https://www.econhist.uni-bayreuth.de

Literature

The following two online textbooks cover many of the tools and concepts covered in this course:

- Robin Lovelace, Jakub Nowosad, Jannes Muenchow (2023). Geocomputation with R. 2nd edition. CRC Press. Available online: https://geocompr.robinlovelace.net/ (short: **LNM**)
- Taro Mieno (2022). R as GIS for Economists. Available online: https://tmieno2.github.io/R-as-GIS for-Economists/ (short: **GIS**)

In addition, we will read and replicate selected journal articles, which I will make available on e-learning. I will also provide more concise resources on e-learning each week, such as useful websites.

Preliminary Schedule

(The schedule may change as we proceed)

Date	Topic	Reading / Lab Activity / Assignments
29.04.	Introduction to the Course	GIS Ch. 1
06.05.	Introduction to R	https://github.com/timuroeztuerk/r-lectures
13.05.	Vector Data I	LNM Ch. 2.2; GIS Ch. 2
20.05.	Vector Data II	LNM Ch. 3.2, 4.2, 5.2; GIS Ch. 3
27.05.	Raster Data I	LNM Ch. 2.3; GIS Ch. 4
03.06.	Raster Data II	LNM Ch. 3.3, 4.3, 5.3; GIS Ch. 5
17.06.	Raster-Vector Interactions	LNM Ch. 6
24.06.	Making Maps with R	LNM Ch. 9
01.07.	Application Session I	TBA
08.07.	Application Session II	TBA
15.07.	Project Presentations I	TBA
22.07.	Project Presentations II	TBA

Office Hours

I will be available for office hours between 12:30 and 14:00 on Tuesdays. Please do visit for any questions you might have, it will bring you a lot!

Academic Integrity

Students are expected to adhere to the University of Bayreuth's standards of academic integrity. I encourage you to collaborate with your peers and use the online tools available. Nevertheless, any submitted work must be your own, uniquely.

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